

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Klastoets 2

Kopiereg voorbehou

Class Test 2

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Module EBN111
Elektrisiteit en Elektronika
13 April 2012

Module EBN122
Electricity and Electronics
13 April 2012



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Toetsinligting:

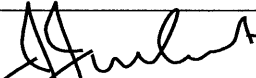
Test information:

Memo

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 minute 30 minutes	Oopboek / toeboek: <i>Open / closed book:</i>	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	8		

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen.
Students may do their own rough calculations on the question paper.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

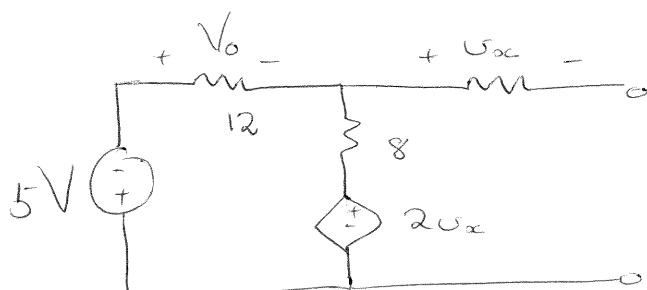
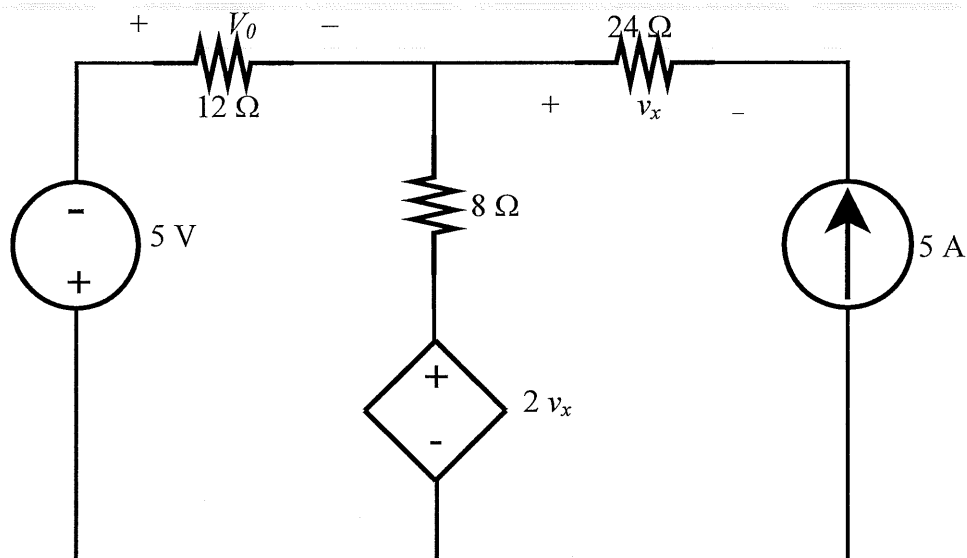
Dosent(e): <i>Lecturer(s):</i>	1 J Joubert		
	2 JD le Roux		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>			
Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	

Vraag 1.1 [2]

Bepaal die spanning V_0 slegs as gevolg van die 5 V spanningsbron.

Question 1.1 [2]

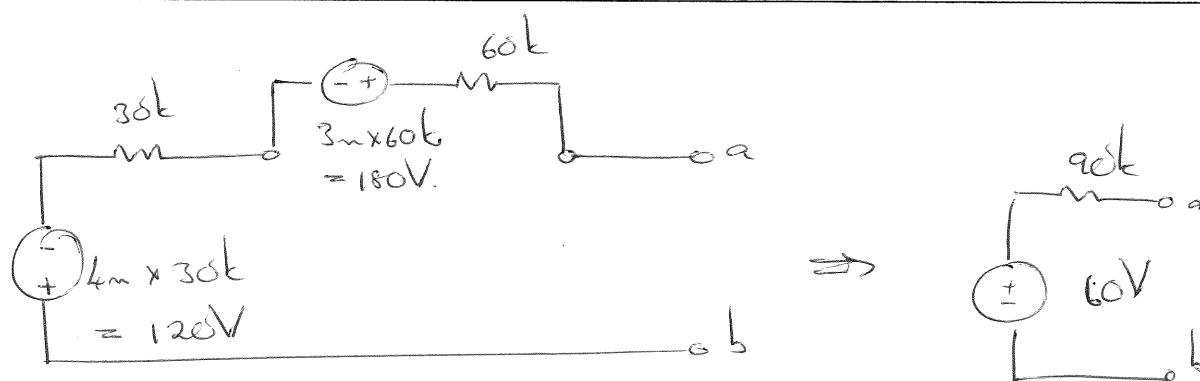
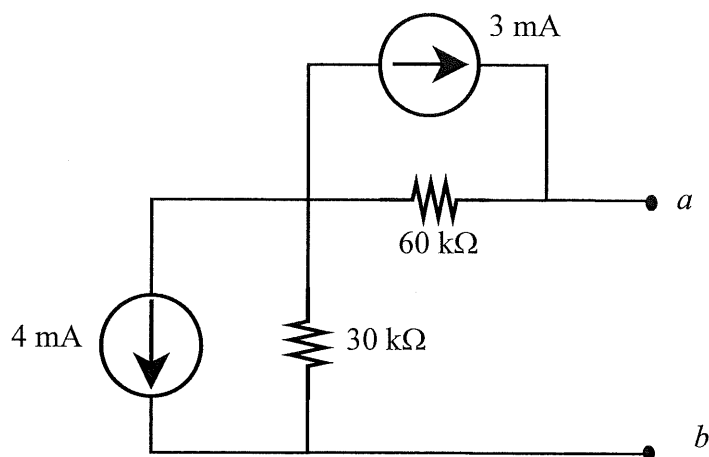
Determine the voltage V_0 only due to the 5 V voltage source.



* $v_x = 0V$ (No current in branch: $v_x = 0 \times 24 = 0V$)

* $V_0 = \frac{-5 \times 12}{12 + 8} = -3V$

(Voltage divider.)

Vraag 1.2 [3]Maak gebruik van brontransformatie om V_{ab} te bereken.**Question 1.2 [3]**Make use of source transformation to determine V_{ab} .

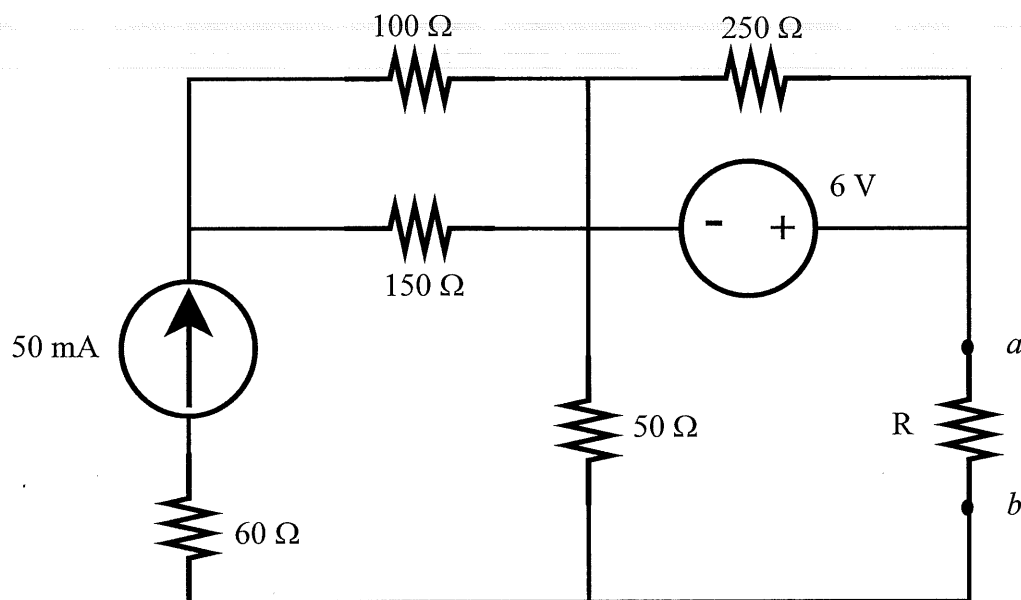
$$\begin{aligned} V_{ab} &= 180 - 120 \\ &= \underline{60\text{ V}} \end{aligned}$$

Vraag 1.3 [4]

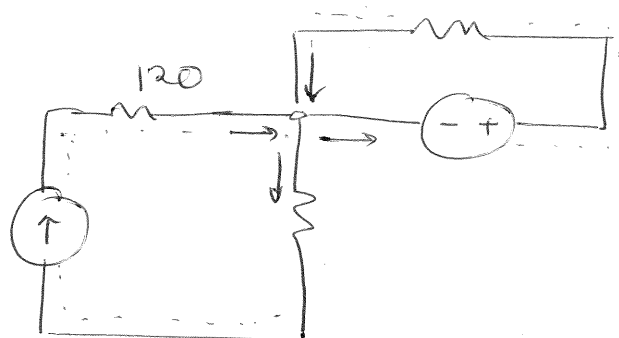
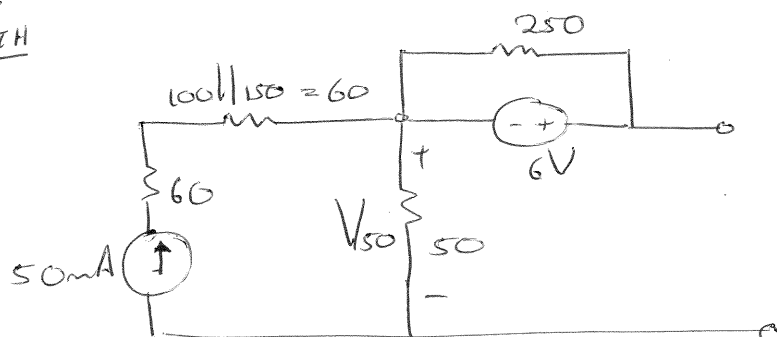
Bepaal V_{TH} en R_{TH} soos gesien deur terminale $a-b$.

Question 1.3 [4]

Determine V_{TH} and R_{TH} as seen by terminals $a-b$.



→ V_{TH}

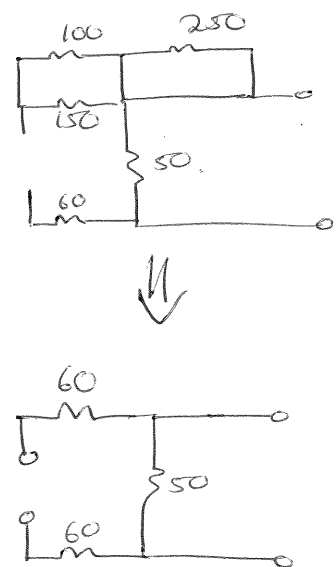


∴ Voltage across 50Ω resistor

$$V_{50} = 50 \times 50 \text{ mA} \\ = 2,5 \text{ V}$$

KVL: $\therefore V_{ab} = 6 + 2,5$
 $= 8,5 \text{ V}$

→ R_{TH}



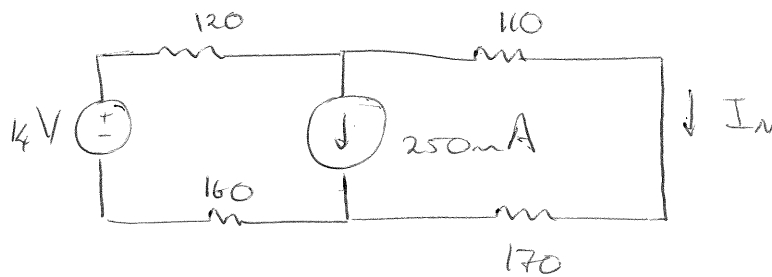
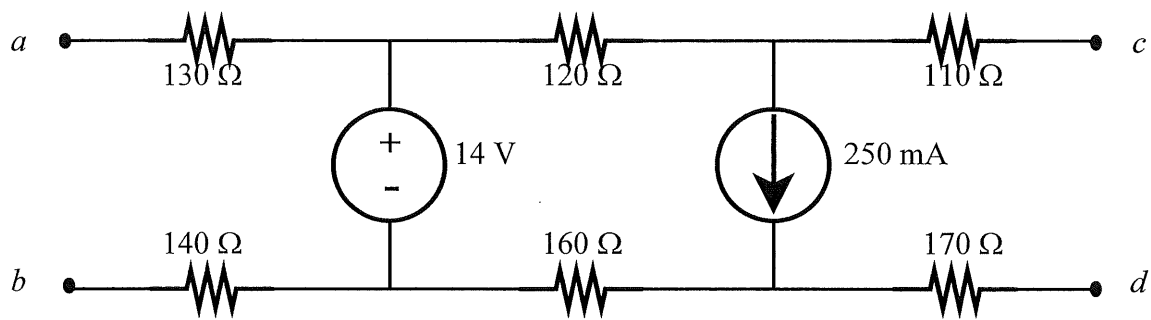
∴ $R_{TH} = 50\Omega$

Vraag 1.4 [2]

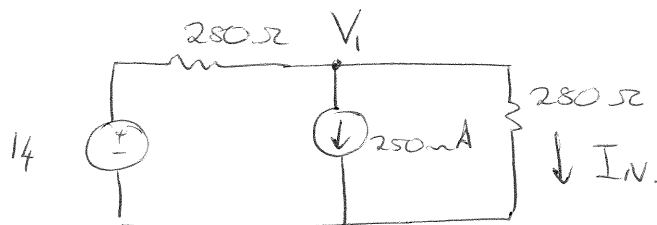
Bepaal I_N soos gesien deur terminale c-d.

Question 1.4 [2]

Determine I_N as seen by terminals c-d.



→ Add series resistors:



→ Nodal:

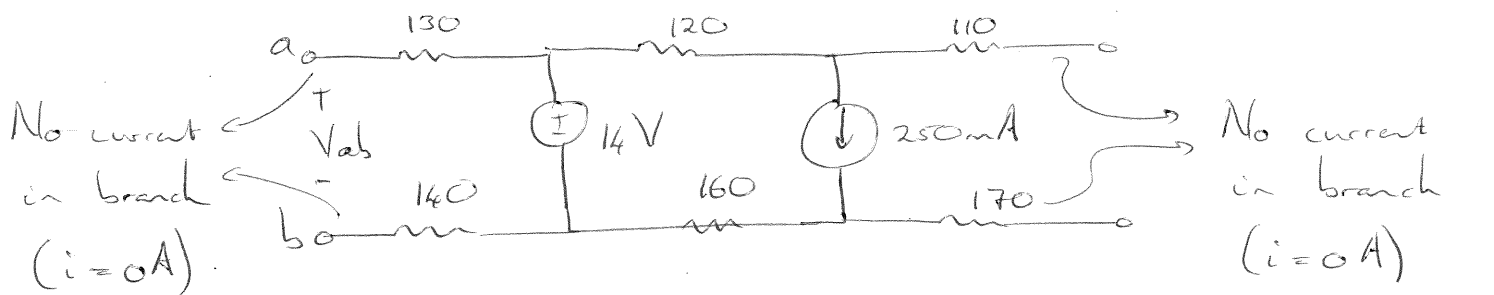
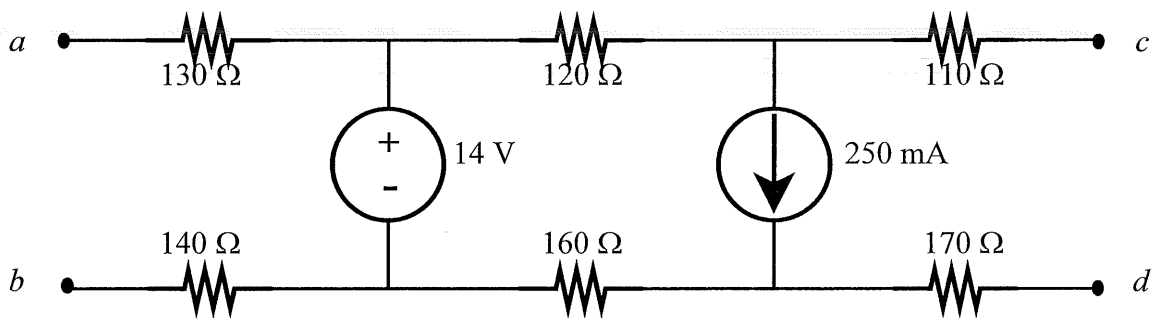
$$\frac{V_1 - 14}{280} + \frac{V_1}{280} = -0,25.$$

$$\frac{2V_1}{280} - \frac{14}{280} = -0,25.$$

$$\frac{V_1}{140} = -0,2$$

$$V_1 = -28V$$

$$I_N = \frac{-28}{280} = -0,1 \text{ A}$$

Vraag 1.5 [2]Bepaal V_{TH} soos gesien deur terminale a-b.**Question 1.5 [2]**Determine V_{TH} as seen by terminals a-b.

$$V_{130\Omega} = 0V$$

$$V_{140\Omega} = 0V$$

$$V_{110\Omega} = 0V$$

$$V_{170\Omega} = 0V$$

$$\begin{aligned} V_{ab} &= V_{130\Omega} + 14 + V_{140\Omega} \\ &= 14V \end{aligned}$$

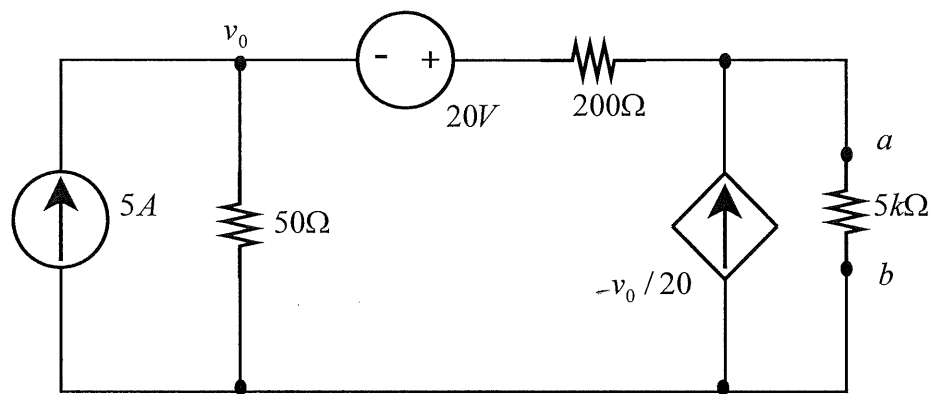
$$\therefore V_{TH} = \underline{14V}$$

Vraag 1.6 [3]

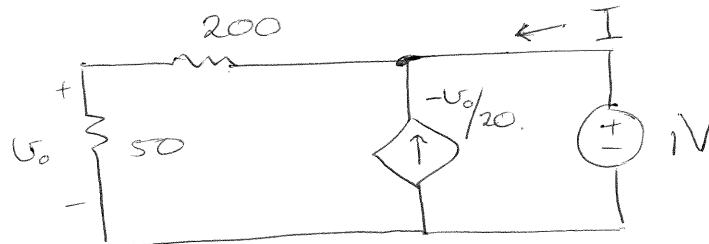
Bepaal R_{TH} soos gesien deur terminale $a-b$.

Question 1.6 [3]

Determine R_{TH} as seen by terminals $a-b$.



R_{TH}



$$* \quad v_0 = \frac{1V}{50 + 200} \times 50 = 0,2V$$

$$* \quad I = \frac{+v_0}{20} + \frac{v_0}{50} = 0,014A$$

$$R_{TH} = \frac{1}{0,014} = \underline{\underline{71,43 \Omega}}$$

EBN111 KLASTOETS / CLASS TEST 2 (13 APRIL 2012)
ANTWOORDBLAD / ANSWER SHEET

Student se besonderhede/Student's details	
Van en voorletters: <i>Surname and initials:</i>	Studentenommer: <i>Student number:</i>
Graadkursus: <i>Degree Course:</i>	
Tel. nr.: <i>Tel no.:</i>	Memo

Tot:	16 /15
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VRAAG 1 – QUESTION 1 (16)	
1.1	$V_0 = -3V$ ✓✓
1.2	$V_{ab} = 60V$ ✓✓✓
1.3	$V_{TH} = 8,5V$ ✓✓ $R_{TH} = 50\Omega$ ✓✓
1.4	$I_N = -9,1A$ ✓✓
1.5	$V_{TH} = 14V$ ✓✓
1.6	$R_{TH} = 71,43\Omega$ ✓✓✓